Arsenous acid reacts with iodine according to the equation below.

$$H_3AsO_3(aq) + I_2(aq) + H_2O(l) \rightarrow H_3AsO_4(aq) + 2\ I^-(aq) + 2\ H^+(aq)$$

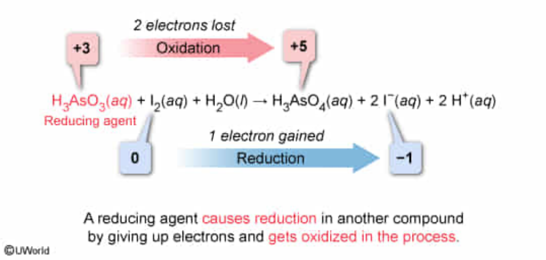
Which chemical species acts as the reducing agent in the reaction?

- A. $H_3AsO_3(aq)$ [59%]
- B. $I_2(aq)$ [11%]
- C. $H_2O(l)$ [24%]
- D. $H_3AsO_4(aq)$ [3%]

Correct
60% answered correctly

Explanation:

Reducing agent in the arsenous acid and iodine reaction



In a redox reaction, oxidation and reduction always occur simultaneously. An atom is reduced by gaining one or more electrons from another atom that loses the electrons (ie, gets oxidized). As a means of **tracking electron transfers** between atoms during reactions, an **oxidation number** can be assessed for each atom in a compound relative to the oxidation state of a neutral atom of that element.

Comparing the oxidation number of an element in the reactants with the oxidation number of that same element in the products indicates whether the element has gained or lost electrons. An *increased* oxidation number indicates a *loss* of electrons (oxidation) whereas a *decreased* oxidation number signals a *gain* of electrons (reduction). A change of **one unit** occurs **for each electron** transferred.

The reducing agent in a reaction is the chemical species that *causes reduction* in another atom by giving up electrons (ie, the reducing agent itself gets oxidized in the process). Conversely, the **oxidizing agent** is the chemical species that *causes oxidation* by taking electrons from (oxidizing) another atom and getting reduced in the process.

In the given reaction, the As atom is oxidized (ie, gains bonds to oxygen), causing the oxidation number of As to change from +3 in $H_3AsO_3(aq)$ to +5 in $H_3AsO_4(aq)$. The electrons lost by the As atom cause $I_2(aq)$ to be reduced, which changes the oxidation number of I from 0 in $I_2(aq)$ (ie, the elemental state) to −1 in the $I^-(aq)$ ion. Therefore, $H_3AsO_3(aq)$ *causes* reduction and is the reducing agent in the reaction.

(Choice B) I_2(*aq*) is the *oxidizing agent* in the reaction.

(Choice C) The oxidation numbers of the H and O atoms do not change during the reaction. As such, the atoms are neither oxidized nor reduced, and H_2O(*l*) cannot be the reducing agent.

(Choice D) H_3AsO_4(*aq*) is a *product* of the reaction and cannot be the reducing agent.

Educational objective:
A reducing agent causes a reduction in another compound by giving electrons to the other compound. Therefore, a reducing agent itself undergoes oxidation due to the loss of electrons. Oxidation numbers track the gain or loss of electrons by atoms in a reaction.

Time spent:0 sec
Subject:
General Chemistry

QID:403962
System:
Interactions of Chemical Substances